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PI Name:	Dingfelder, Michael Ph.D.		
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PI Email:	dingfelderm@ecu.edu	**Fax:**	FY 252-328-6314
PI Organization Type:	UNIVERSITY	**Phone:**	252-328-0882
Organization Name:	East Carolina University		
PI Address 1:	Department of Physics		
PI Address 2:	Mailstop 563		
PI Web Page:			
City:	Greenville	**State:**	NC
Zip Code:	27858	**Congressional District:**	3
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):	Toburen, Larry (East Carolina University)		
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Task Description:

The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment.

The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells.

Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Work is currently in progress in several directions. We are working on the development and improvement of the PARTRAC module for the simulation of the HZE particles. This includes an update of the liquid water cross sections and the development of interaction cross sections for calcium, a major constituent of bone. Work is also in progress to develop and test realistic bone/bone-marrow target models using the geometry package of PENELOPE. This includes the development of a generic model which can be used for different species (humans, rodents) and different bone sites. We have tested this model for two specific human bone sites, a lumbar and a thoracic vertebra. Currently we are working on extending the model to rodents. We are also working on the coupling of the geometry model (PENELOPE) with the HZE track structure modules of PARTRAC.
We have developed an interaction model for bare (i.e., fully ionized) and sufficiently fast heavy ions by relating HZE interaction cross sections to ionization and excitation cross sections of protons by scaling laws. HZE inelastic interaction cross sections (total and energy differential) are obtained from the proton cross sections by energy scaling and multiplication by the ion charge squared. Secondary electron emission angles are modeled as a superposition of soft collisions (dipole interaction) and hard or binary collisions; they are calculated for each ion separately.

The current HZE transport module is based on the proton ionization and excitation cross sections with liquid water as implemented in PARTRAC. Cross sections in the original PARTRAC are calculated within the non-relativistic plane-wave Born approximation and the relativistic Bethe approximation. Track structure simulations at higher initial energies resulted in lower than expected energy losses (or LET values), i.e., current simulations yield a LET of 110 keV/micron for a 1 GeV/nucleon 56Fe26+ instead of the experimentally observed 150 keV/micron. We are currently recalculating proton interaction cross sections using the old model for the dielectric response function together with a fully relativistic plane-wave Born approximation (RPWBA) including condensed phase effects like the Fermi-density effect. These calculations are numerically delicate and computationally expensive. In contrast to the larger discrepancies in the differential cross sections as observed earlier, calculated stopping powers agree better with the current ICRU recommendations. Further details on the current status of the track structure codes are published in M. Dingfelder, Radiat. Prot. Dosim. (2006) in press; published online doi:10.1093/rpd/ncl494 and M. Dingfelder et al., Radiat. Prot. Dosim. (2006) in press; published online doi:10.1093/rpd/ncl415.

We also made progress in our goal to obtain interaction cross sections for HZE particles with calcium, a major component of bone. We collected a variety of experimental and theoretical data on optical properties and the dielectric response of (metallic) calcium in the literature and modeled the extinction coefficient and dielectric response of calcium. At the moment we are adjusting our model parameter to comply with theoretical sum rules and other properties like the mean excitation potential.

We have developed a simple, non-bone-site-specific, non-patient-specific, and computationally efficient geometrical model of trabecular spongiosa using the geometry package of the general-purpose MC transport code PENELOPE. Internal dimensions of the model are obtained from path-length distributions as measured in different human bones by research groups at the University of Florida and the University of Leeds. We have benchmarked and validated our

	model by determining path-length distributions and by calculating absorbed dose by internal monoenergetic electron-emitters and by comparing it to more sophisticated tomography-based and chord-length-based transport models published in the literature. Details of the model are published in J.A. Gersh, M. Dingfelder, and L.H. Toburen, Modeling energy deposition in trabecular spongiosa using the Monte Carlo code PENELOPE, Health Physics (2007) accepted. We are currently gathering information on rodent bones and evaluating ways to obtain internal dimension parameters needed for our bone/bone-marrow model. We have established collaboration with Dr. Derek Jokisch from Francis Marion University on further developing and testing the human bone model and helping gathering input data for the rodent model, i.e., analyzing micro-CT scans. Dr Jokisch is part of the University of Florida group that has an extensive history in bone modeling in humans. Also, we are currently establishing contacts to Dr. Ted Bateman from Clemson University to obtain micro-CT scans of rodent bones. We are currently working on coupling the event-by-event code PARTRAC – which supports only slab-geometries – with our trabecular spongiosa geometry model. As a first step we plan to transport HZE particles with PARTRAC while secondary electrons are transported with PENELOPE. We have developed a method to reduce our realistic geometry used in PENELOPE to a simple slab geometry which can be used in PARTRAC. Currently, we are testing the hybrid code of both systems transporting alpha particles from external as well as internal emitters.
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